

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include "commonType.h"
5  #include "initAndTerm.h"
6  #include "dataControl.h"
7  #include "fileLoadWrite.h"
8
9  extern node *head, *tail;
10
11 int exeFunc(int (**pF)(void), int sel, int *ret){
12     pF[sel]();
13     *ret = 0;
14     return 0;
15 }
16
17 int insertNode(node *head, node *tail, elem data){
18     elem *newNode, *list;
19     list = (elem *)head;
20
21     while(list->next != (elem *)tail){
22         list = list->next;
23         if(list->num == data.num){
24             printf("입력 실패 했습니다.(같은 번호존재)\n");
25             return 0;
26         }
27     }
28     newNode = (elem *)malloc(sizeof(elem));
29     memset(newNode, '\0', sizeof(elem));
30
31     newNode->prev = list;
32     newNode->next = (elem *)tail;
33     strncpy(newNode->name, data.name, sizeof(newNode->name));
34     newNode->age = data.age;
35     newNode->num = data.num;
36     head->cnt++;
37     tail->cnt++;
38
39     list->next = newNode;
40     tail->prev = newNode;
41     return 0;
42 }
43
44 int deleteNode(node *head, node *tail, int num){
45     elem *list;
46     list = head->next;
47     while(list != (elem *)tail){
48         if(list->num == num){
49             list->next->prev = list->prev;
50             list->prev->next = list->next;
51             head->cnt = head->cnt-1;
52             tail->cnt = tail->cnt-1;
53             free(list);
54             return 0;
55         }
56         list = list->next;
57     }
58     printf("삭제할 번호가 없습니다.\n");
59     return 0;
60 }
61
62 int printNode(node *head, node *tail){
63     elem *list;
64     list = head->next;
65     printf("|번호|이름|나이|\n");
66     while(list != (elem *)tail){
67         printf("|%d|%s|%d|\n",
68             list->num, list->name, list->age);
69         list = list->next;
70     }
71     return 0;
72 }
73
74 int selectInsert(){
75     elem data;
76     printf("번호를 입력하세요: ");
77     if(scanf("%d", &data.num) == 0){
78     }
79     while(getchar() != '\n');
80
81     printf("이름을 입력하세요 (제한글자수 %d): ",
82         (int)sizeof(data.name)-1);
83     if(fgets(data.name, sizeof(data.name), stdin) == 0){
84     }
85     if(data.name[strlen(data.name)-1] == '\n'){
86         data.name[strlen(data.name)-1] = '\0';
87     }
88     fflush(stdin);
89
90     printf("나이를 입력하세요: ");
91     if(scanf("%d", &data.age) == 0){
92     }
93     while(getchar() != '\n');
94     insertNode(head, tail, data);
95
96     return 0;
97 }
98
99 int selectDelete(){
100     int num;
101     printf("번호를 입력하세요: ");
102     if(scanf("%d", &num) == 0){
103     }
104     deleteNode(head, tail, num);
105     return 0;
106 }
107
108 int selectPrint(){
109     printNode(head, tail);
110     return 0;
111 }
112
113 int selectEnd(){
114     saveData(head, tail);
115     releaseNode(head, tail);
116     return 0;
117 }
118
```